



Sparse Reward Benchmark for Reinforcement Learning Exploration

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Abstract

Sparse rewards make reinforcement learning difficult by providing limited training feedback. We present a sparse-reward benchmark to evaluate exploration strategies, comparing Entropy-Regularized PPO with Intrinsic Curiosity Module (ICM) and Random Network Distillation (RND) across control and robotic manipulation tasks. Results show that intrinsic rewards improve exploration coverage, but greater exploration does not always translate into faster learning or higher success rates. The effectiveness of exploration is also influenced by the reward structure, highlighting a trade-off between exploration diversity and policy optimization.

Research Objectives

- **Objective 1:** Develop a sparse-reward benchmark that enables systematic evaluation of exploration strategies across environments with different state and action spaces.
- **Objective 2:** Compare Entropy-regularized PPO, ICM, and RND in terms of learning performance and exploration behavior under sparse-reward settings.
- **Objective 3:** Investigate the relationship between exploration coverage and task success, and analyze how reward design influences learning efficiency.

Experimental Setup

Category	Setup
Environments	GridWorld, FrozenLake, Sparse CartPole, Reach-v3, Button-Press-v3
Methods	Entropy-regularized PPO, Entropy-regularized PPO+ICM, Entropy-regularized PPO+RND
Metrics	Success Rate, Average Return, EWMA Success Rate, Heatmaps

Results: Gridworld

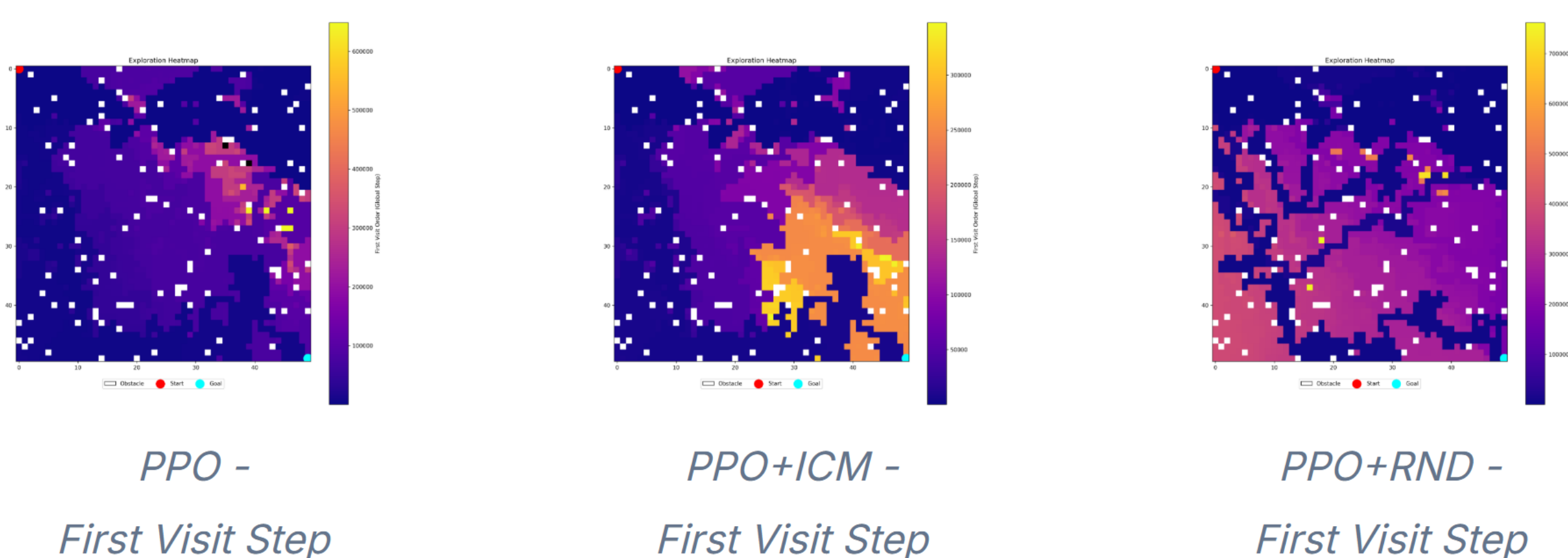


Figure 1. Gridworld first visit step heatmaps.

Observation In a 50×50 environment with 100 obstacles, PPO learns the fastest, but it does not fully explore the whole map, leaving a few scattered cells unvisited. ICM is a bit slower, but achieves complete exploration of the entire map thanks to curiosity-driven rewards. Meanwhile, RND initially clears a few clear and distinct paths, but it ends up being the slowest to learn because the external sparse reward wasn't strong enough to beat the intrinsic curiosity reward.

Results: FrozenLake

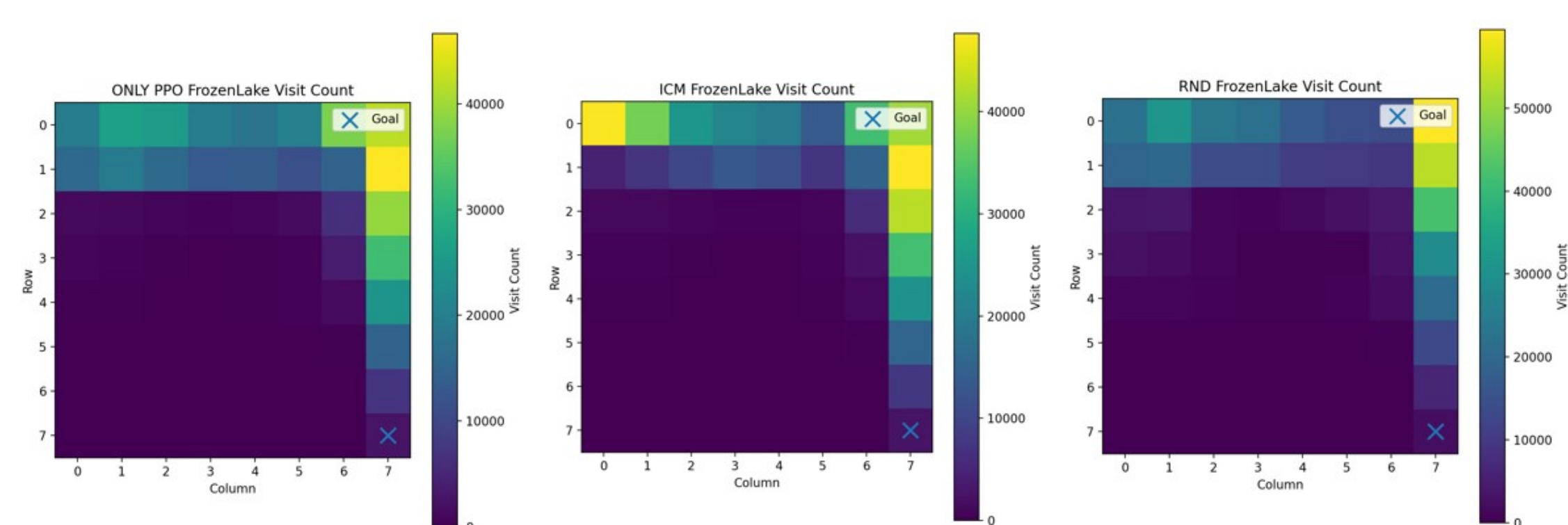


Figure 2. FrozenLake state visitation heatmaps.

Observation ICM begins learning earlier than PPO and RND, demonstrating the benefit of curiosity-driven exploration during early training. Although PPO, ICM, and RND exhibit different state visitation patterns, with ICM showing stronger local exploration around the starting region, all methods eventually converge to similar task performance on FrozenLake.

Results: Sparse CartPole

Task Modified CartPole-v1: reward +10 only when the cart crosses $x = 1.5$ with the pole upright; 0 otherwise. The agent must balance *and* drive toward the goal.

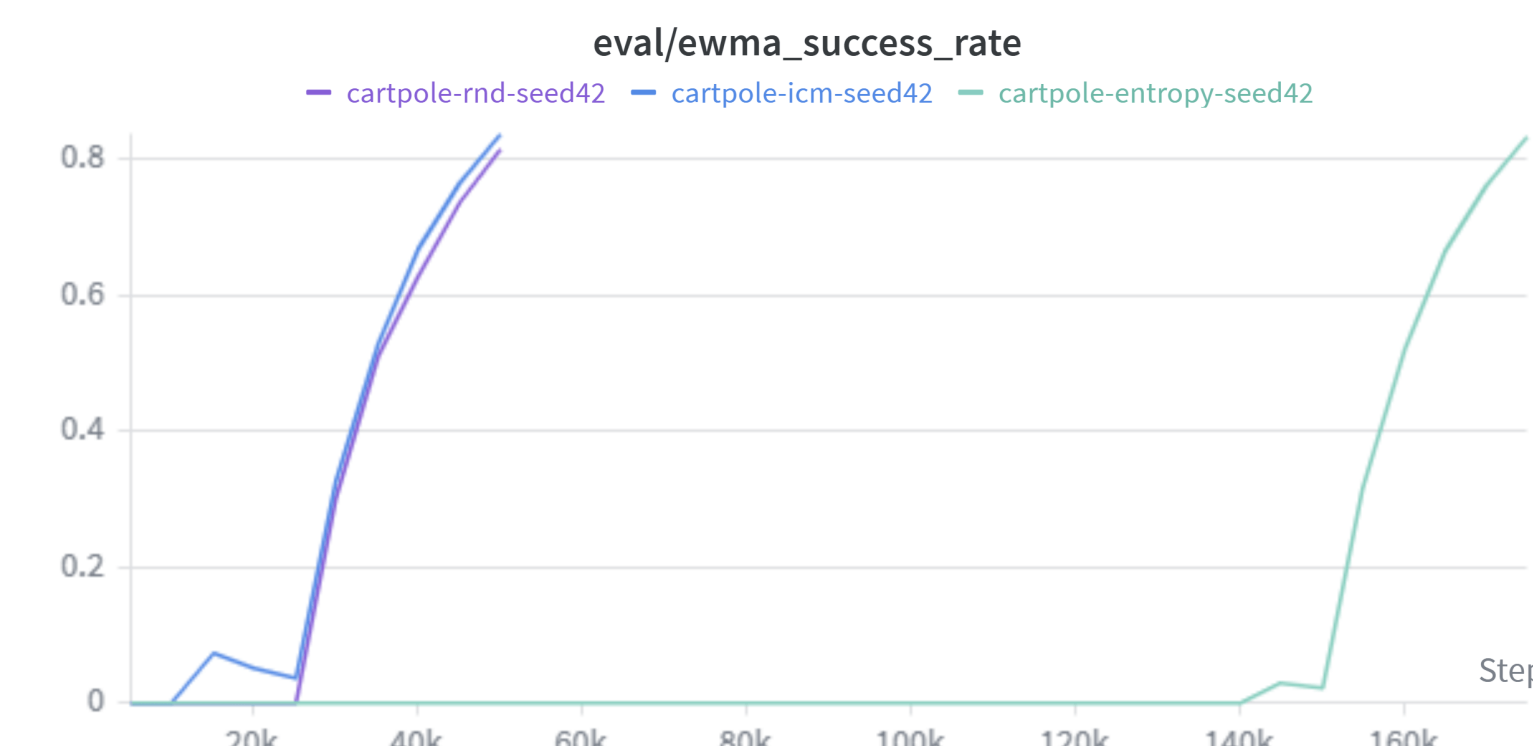


Figure 3. EWMA success rate on Sparse CartPole.

Observation Both PPO+ICM and PPO+RND substantially accelerate goal discovery compared to the PPO baseline, performing almost identically to each other. However, all three methods eventually converge to the same final success rate, indicating that intrinsic motivation benefits exploration speed rather than final task performance.

Results: Meta-World Reach-v3

- Dense shaping (0.005) substantially improves learnability.
- PPO performs best with the basic reward wrapper.
- ICM and RND benefit more from milestone-based rewards.
- Reward structure strongly influences exploration effectiveness.
- Better exploration does not always lead to better task performance.

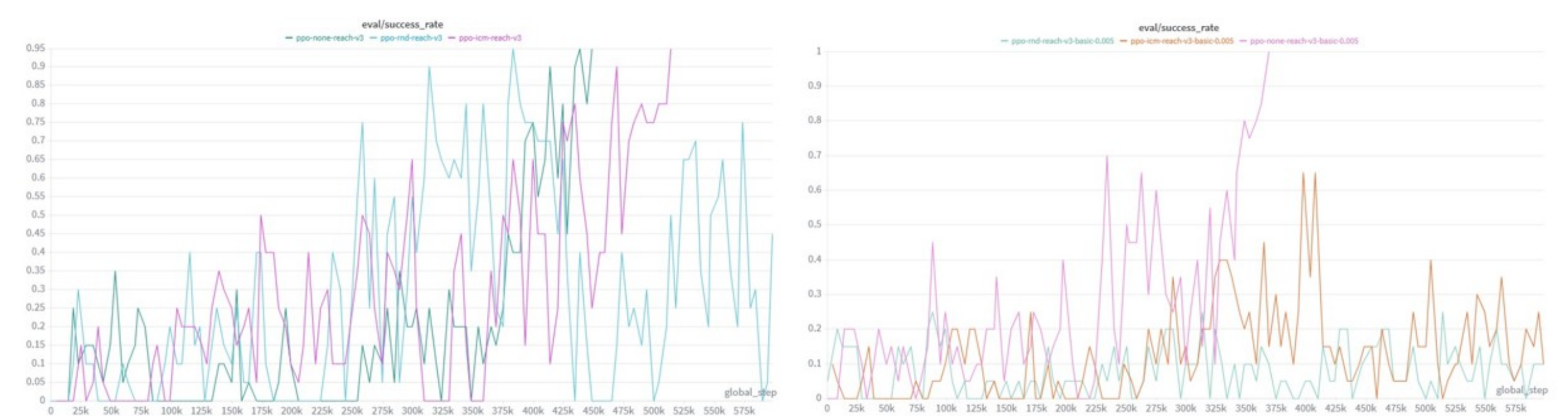


Figure 4. Comparison between milestone and basic wrappers with 0.005 reward shaping.

Results: Meta-World Button-Press-v3

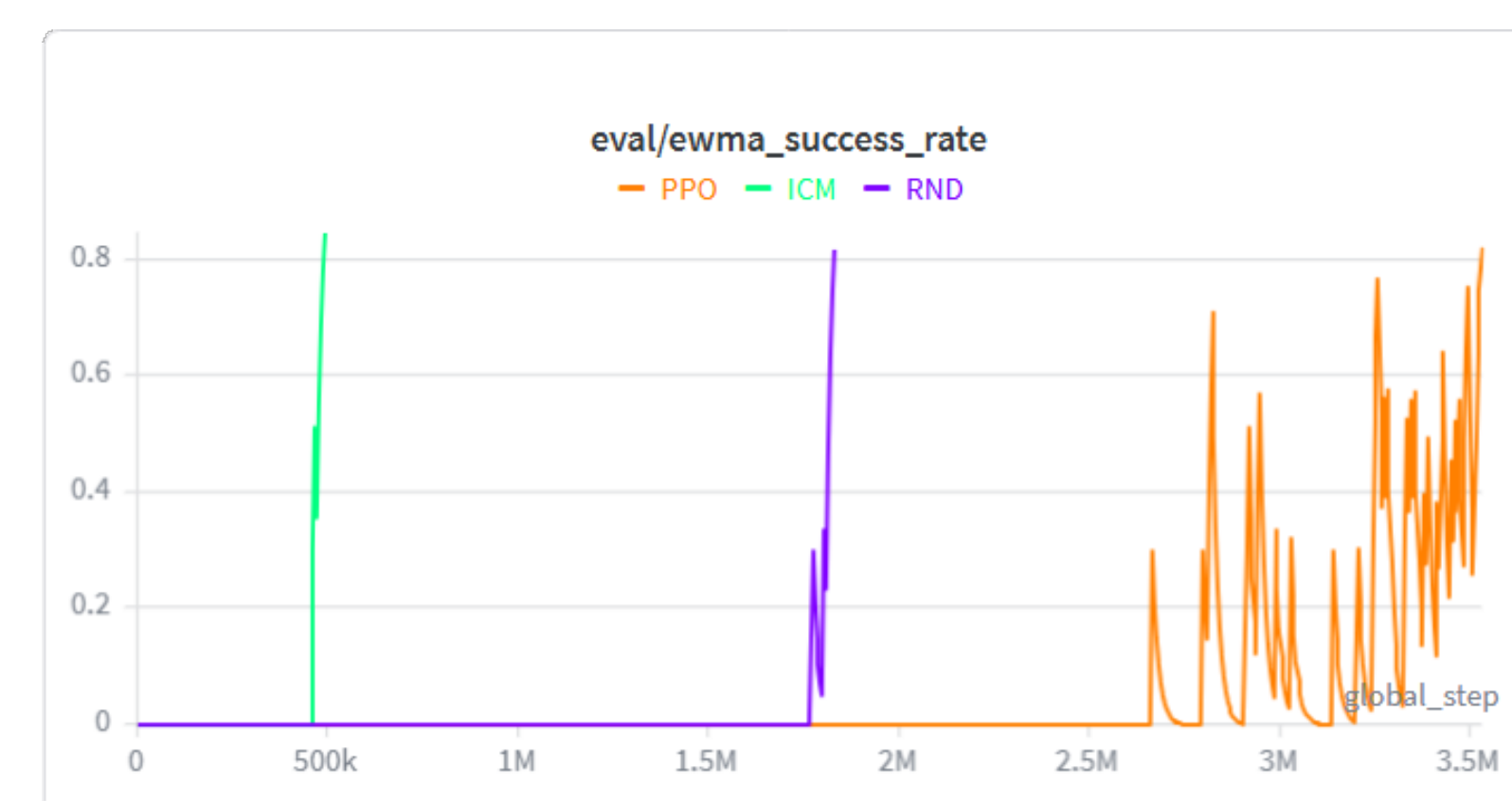


Figure 5. EWMA success rate on Meta-World Button-Press-v3.

Observation All methods experience a long initial period with a zero success rate. However, ICM is the first to break through and shoot up. This may be because button-pressing involves dynamic physical interaction; once the button is accidentally pressed, the sudden state change generates a strong intrinsic curiosity reward. This quickly solidifies the successful policy. In contrast, RND is delayed by getting stuck in corners, and standard PPO lacks the active drive to escape its early movement patterns, making them slower to succeed.

Conclusions

- We developed a unified sparse-reward benchmark across discrete and continuous-control tasks.
- No single exploration method consistently performs best across all environments.
- Performance depends strongly on task type and reward structure.
- Sparse-reward exploration should be evaluated through the interaction between environment, reward design, and exploration mechanism.